Straight-line graphs — Plotting · Gradient · $y = mx + c$

Squared paper needed. ★ = challenge question.

1 Draw axes from -6 to 6. Draw the graph of $y = 2x - 3$ using a table of values for $x = -1, 0, 1, 2, 3$.

2 Draw the graph of $3x + 2y = 6$.

3 Four lines: $y = x$, $y = 3x$, $y = 2x + 4$, $y = 2x - 1$.

- **a** Which lines pass through the origin?
- **b** Which line is the steepest?
- **c** Which two lines are parallel?
- **d** Which line crosses the y-axis lowest down?

4 Work out the gradient of the line through each pair of points.

a $(0, 1)$ and $(2, 5)$ **b** $(-2, 3)$ and $(4, 0)$ **c** $(1, -4)$ and $(3, -4)$

5 A line passes through $(0, -2)$ and $(4, 6)$. Write down its equation.

6

- **a** Where does the line $y = 3x - 5$ cross the y-axis?
- **b** What is the gradient of the line $y = -2x + 7$?
- **c** Write the equation of the line with gradient 4 that crosses the y-axis at $(0, -3)$.

7 Does the point $(6, 2)$ lie on the line $y = 2x - 14$? Show working to explain.

8 Write the equation of each line.

- **a** Through $(0, 3)$ and $(3, 0)$.
- **b** Through $(0, 2)$ and $(1, 5)$.
- **c** Through $(0, -3)$ and $(4, -1)$.

9 A line $y = mx + c$ passes through the point $(1, 3)$. Write down three possible equations for the line. What is the relationship between m and c?

10 Line A is $y = 2x + 1$. Line B is parallel to A and passes through $(3, 2)$. Line C is $y = -\frac{1}{2}x + 1$.

- **a** Find the equation of line B.
- **b** Where do lines A and C cross?

Check: $y = mx + c - m$ is the gradient, c where it crosses the y-axis · parallel lines have the same m · a point is on the line only if its coordinates satisfy the equation.